



**NUS Orbital 2024
Final Submission**

Crossroads by Team Calamari & Basil

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POSTERS

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PROBLEM STATEMENT

Modern social media apps facilitate interactions and connections, but most of them emerge to be ingenuine/ inorganic.

WHY CROSSROADS?

CrossRoads redefines the way people meet and connect in the digital age. We heavily emphasise spontaneity, facilitating meaningful and organic interactions between users in the real world.

SWE PRACTICES

SOLID principles:
Designing software that is understandable, flexible, and maintainable

Ensuring entities are open for extension but closed for modification
(Open/Closed principle)

Branching:
Concurrent development and risk reduction; ensuring more streamlined code review processes

EVALUATION

1. Continuous Integration/Continuous Delivery
2. Unit Testing
3. Manual System Testing
4. User Testing



KEY FEATURES

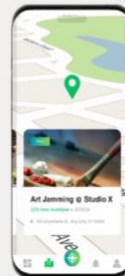
Location-based bumps

Utilises advanced geolocation technology to recommend potential matches based on live location. Users are mainly allowed to interact with one another based on their proximity, provided they turn on this feature.



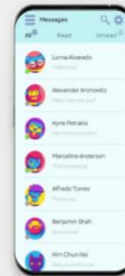
Activity Hub

Allows users to either advertise their activities (once approved) or discover exciting events and gatherings to join.



Real-time messaging

Supports real-time messaging between users to enable people to determine if, where and when they should meet up after connecting.



TECH STACK



Cross Paths, Start Sparks ✨

Team Name

Calamari & Basil

Proposed level of achievement

Apollo 11

Motivation

We undertook the Orbital project due to our passion and curiosity for software development and our desire to immerse ourselves in a self-directed learning journey.

With the opportunity to engage in independent work during the summer break, we aim to harness the resources available through as well as the internet Orbital to expand our skillsets through hands-on work, developing a product which we are passionate about.

Aim

Our goal for Orbital is to conceptualise, develop, and deliver a social media mobile application that addresses a prominent social problem of a lack of meaningful social interactions. The irony of our digital world is that people have never been more interconnected with each other through phone calls or messages and yet many still feel lonely due to the lack of meaningful social connections with one another.

Our social media application “CrossRoads” aims to address this problem by becoming the platform that allows people to meet and engage with one another as spontaneously as possible through geolocation technology. Users near each other are discoverable and allow users to get to know people in the moment without the loathing thought of being judged for approaching someone.

Our mobile application offers a safe space for people to meet, chat and connect with one another in real-time. It even extends to social activities such as birthday parties, hiking trips, exercise events, etc. Given the complex nature of this social media

platform, our team would like to challenge ourselves by pushing our boundaries and embarking on the Apollo 11 Orbital project so that we will be able to hone our software engineering skills better and become well-rounded individuals equipped with relevant skill sets applicable in addressing prominent problems of the world.

User Stories

“CrossRoads” Social Media Mobile Application: Get to know & meet new people based on live location (enables spontaneity & creates an organic meeting experience)

CrossRoads is a revolutionary social media app that redefines the way people meet and connect in the digital age. Emphasising spontaneity and organic interactions, CrossRoads leverages real-time location data to facilitate meaningful encounters, foster new relationships, and create lasting memories.

1. As a person who is single and wants to be in a relationship, I want to be able to meet new people by messaging people who have crossed paths with me.
2. I wish to meet people more organically, which modern dating apps just seem unable to provide.
3. As a person who wants to attend social events but does not know where to start, I can indicate my interest/attendance at a social event listed on the mobile platform.
4. As an introvert, I would seldom approach people in public even if I have the urge to. CrossRoads would make such interactions less intimidating.
5. As a frequent social media user, I want to be able to post and consume media content on the mobile platform. I want to connect with my matches on a deeper level and understand them better
6. As a dating app user, I might feel unsafe meeting strangers individually sometimes. Having the opportunity to meet people in groups can provide a sense of security.

Development Plan

Week 1 - 2 (May 13 - June 03, 2024)

1. Milestone 1 - Technical proof of concept (i.e., a minimal working system with both the frontend and the backend integrated for a very simple feature)
 - a. Login/Sign up page
 - b. Real-time messaging should be completed between at least 2 users
 - c. Notifications prompt when the user is within the specified radius

Week 3 - 5 (June 04 - June 30, 2024)

2. Milestone 2 - Prototype (i.e., a working system with the core features)
 - a. Location-based recommendations should feature multiple profiles within the specified radius
 - b. Real-time messaging between multiple users
 - c. Notifications prompt when the user is within the specified radius
 - d. ActivityHub allows the listing of an event & allows attendance of multiple events
 - e. Allows Social Media Content browsing & upload

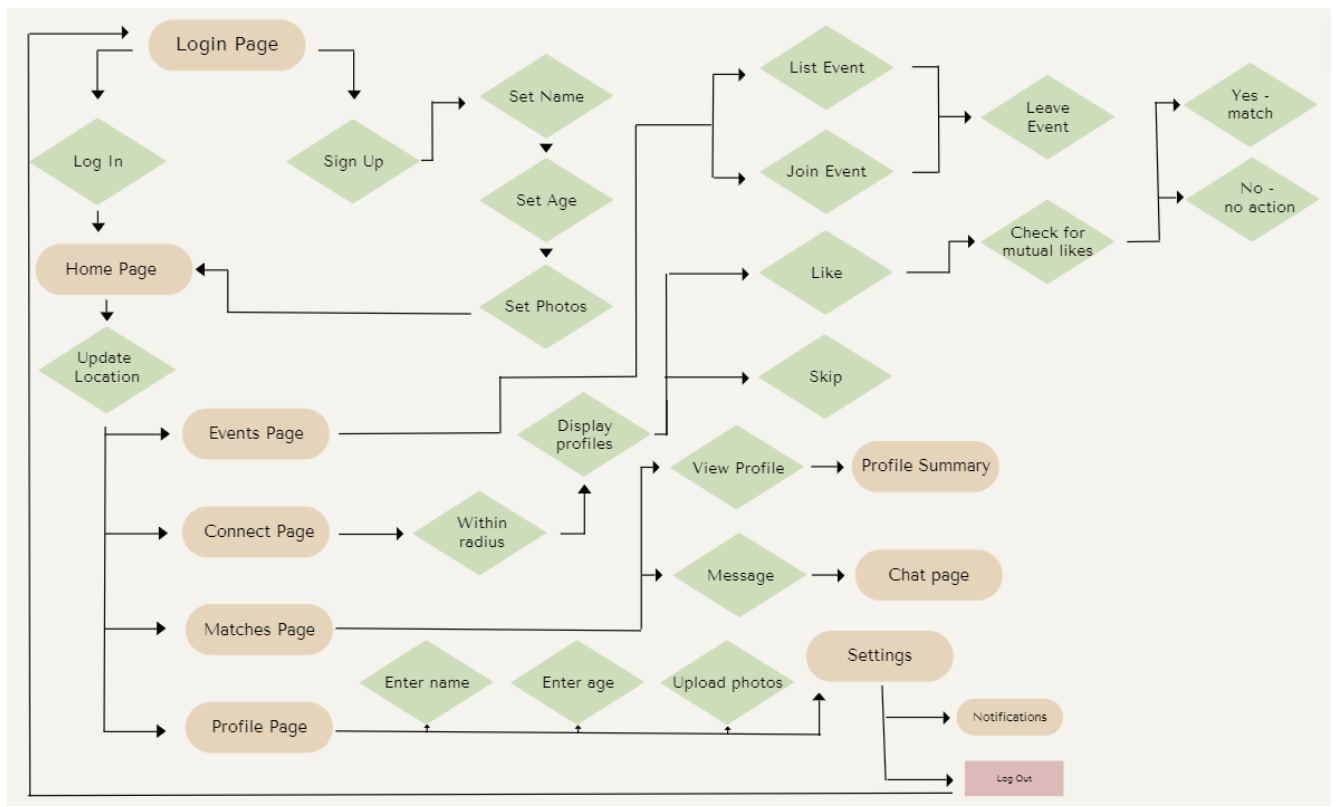
Week 6 - 8 (1 July - July 8, 2024)

3. Milestone 3 - Extended system (i.e., a working system with both the core + extension features)
 - a. Location-based recommendations should feature multiple profiles within the specified radius
 - b. Real-time messaging between multiple users
 - c. Notifications prompt when the user within the specified radius
 - d. ActivityHub allows the listing of an event & allows attendance of multiple events
 - e. Allows Social Media Content browsing & upload
 - f. Has Points system integrated

Wireframe

<https://www.figma.com/design/xvSAQxg0sGREIJAkP9fPSy/CrossRoads-Project?node-id=0-1&t=rw2xYYM1ld9orRu1-1>

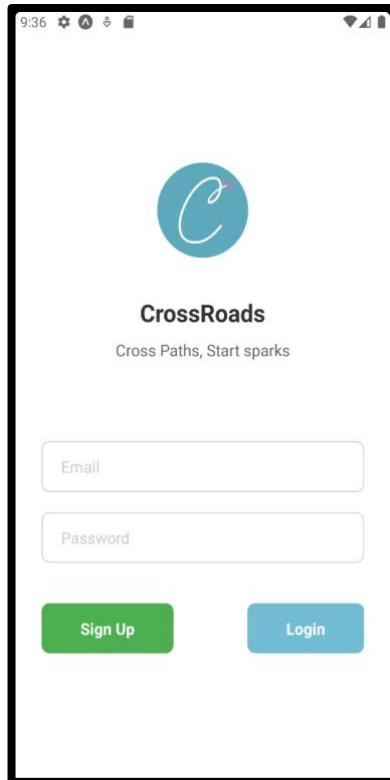
Overall Navigation Flow



Features

1. **User Account Authentication** [Completed]

Utilising the Firebase SDK Authentication, users can sign up for CrossRoads using their emails, and log in with the password that they set.



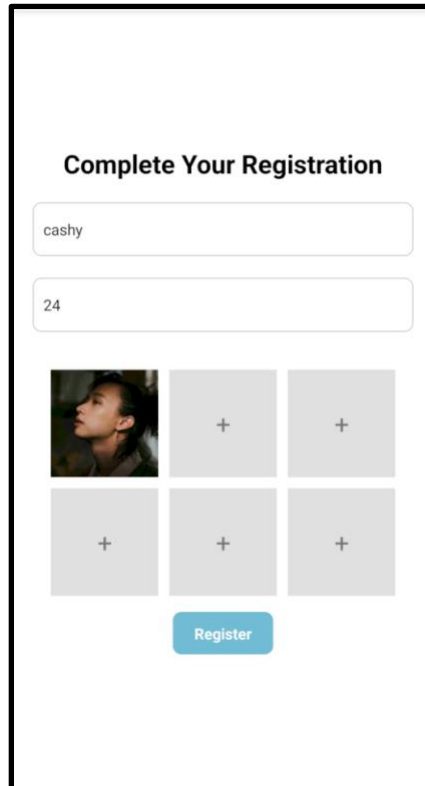
Sign-in method: Email and Password

Based on the user's action, the relevant call to the Firebase authentication instance is made. E.g. registering with email and password calls `createUserWithEmailAndPassword()`.

The `FirebaseAuthExceptions` are handled within a try-catch block in the case of invalid emails, invalid passwords, or trying to create an already existing account etc. The user will be directed to the home page once done with registration/ logging in.

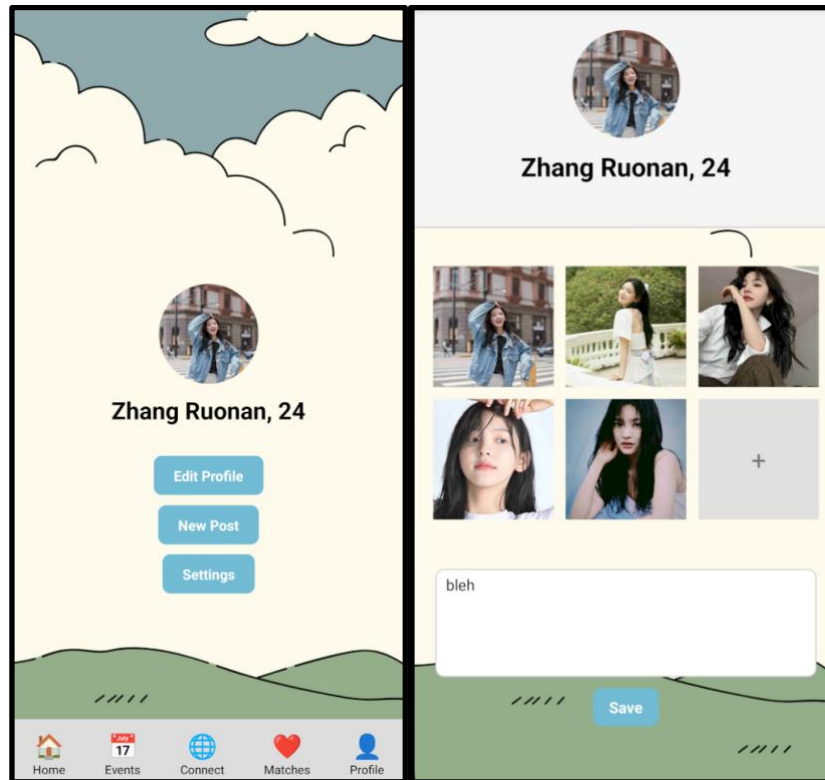
2. Registration & Profile setup **[Completed]**

To finish setting up their accounts, users are required to attach photos of themselves along with their name and age. This information will be publicly available to other potential matches through the **Connect** page.



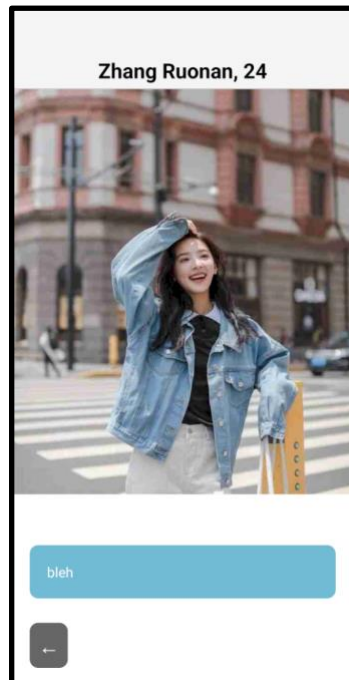
The screenshot shows a mobile app interface for completing registration. At the top, the title "Complete Your Registration" is centered. Below it are two input fields: the first contains the text "cashy" and the second contains "24". Under the input fields is a 2x3 grid of image placeholders. The top-left placeholder shows a profile picture of a woman, while the other five placeholders are gray squares with a white plus sign. At the bottom center of the grid is a blue button with the text "Register".

Thereafter, the user can make changes to their pictures/ user description through the **Edit Profile** component of the **Profile** page.



Other users will be able to view the main profile picture that is uploaded by the user on the “connect” page. The user can re-upload new photos at any point in time which will be updated in the Firebase Storage and then retrieved as part of user profile information.

Additionally, after matching, users will be able to see the full profile of their matches by clicking on their names in the **Matches** page.



This page enables navigation through tap gestures; tapping the left half goes to the previous image and the right half goes to the next image of that match, stopping at the first and last image(s).

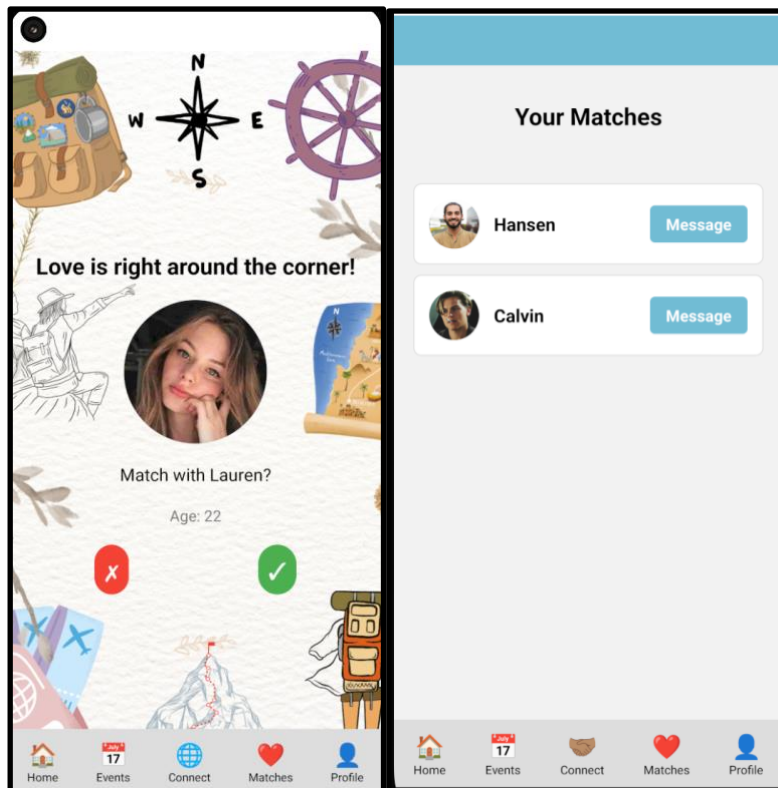
3. Location-Based Recommendations (Core) [Completed]

CrossRoads utilises geolocation technology to recommend potential matches based on live location. Users receive real-time suggestions for nearby people and activities, enhancing the chances of spontaneous meet-ups. This is implemented in the **Connect** matching system as part of our matching algorithm. We are utilise the expo-location and geofire-common libraries to get the user's current location and generate a geohash for distance calculation purposes.

Once a user logs in, they will be prompted to share their location data with the CrossRoads application. This location is updated in Firestore everytime the user logs in, and profiles are then filtered based on whether users are within a 5km radius of each other; if a user is within a 5km radius of another, they will appear in each others' **Connect** page.

4. Connect (Core) [Completed]

Without a functional and well-designed matching system, no dating app can be considered functional. CrossRoads deploys a simple and user-friendly interface for users to view the profiles of potential matches and decide whether or not to connect with them.

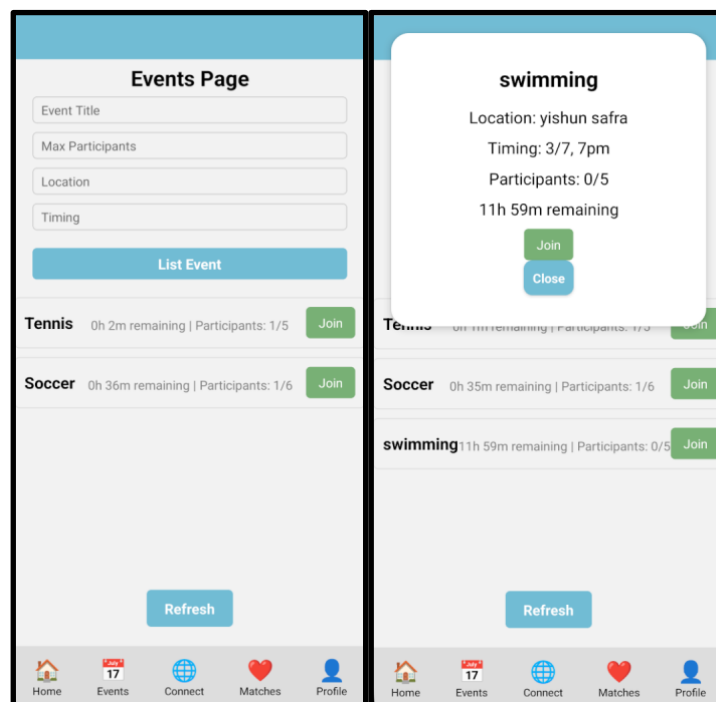


When one user likes another (after being in proximity of each other), this data is stored as a collection in the Firestore Database.

If there exist mutual likes, the IDs of both users are then saved in another collection of matches, and user data will then be retrieved from the Matches page and displayed on the screens of both users.

5. Activity Hub (Core) [Completed]

In the bustling world of CrossRoads, spontaneity isn't limited to just meeting new people; it extends to activities and experiences. The Activity Hub is a unique feature that allows users to either advertise their activities or discover exciting events and gatherings to join, all of which have a time limit of just 1 hour. This time constraint ensures that users seize the moment and engage in real-time, creating an atmosphere of immediate excitement and adventure.

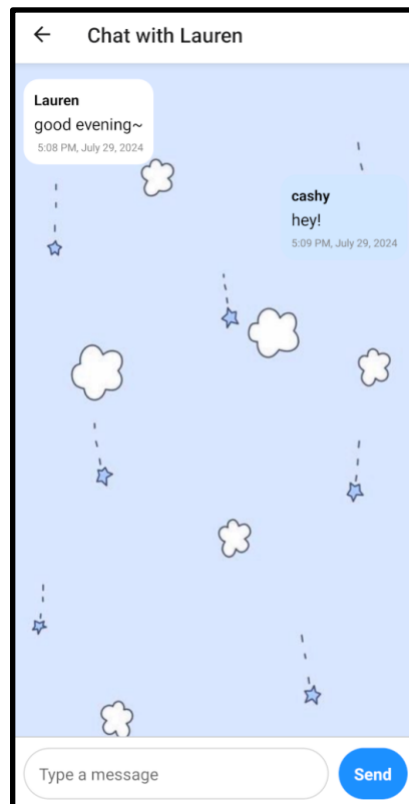


Currently, users can view all listed events, their cut-off timing for signups, number of participants, location as well as timing.

Future plans include allowing users to filter these activities by interest, and location. We wish to display more details regarding the event (including AI generated images for better UX) as well as create group chats for these events.

6. Real-time messaging (Core) [Completed]

CrossRoads supports real-time messaging between users to enable people to determine if, where and when they should meet up so that organic connections can be made as spontaneously as possible.



This component utilises Firebase for authentication and Firestore for real-time message storage and retrieval; it adds a new document to the 'Messages' collection in Firestore with the message text, current timestamp, user ID, user name, recipient ID, and recipient name.

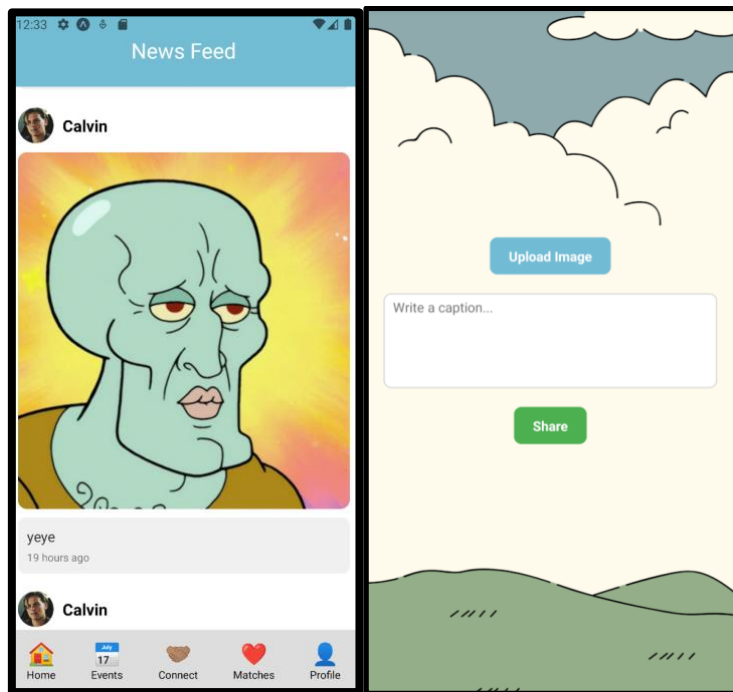
If the user is authenticated, a Firestore query is set up to listen for messages between the current user and the recipient, ordered in ascending order.

The onSnapshot function listens for real-time updates to the query and updates the messages state with the latest messages.

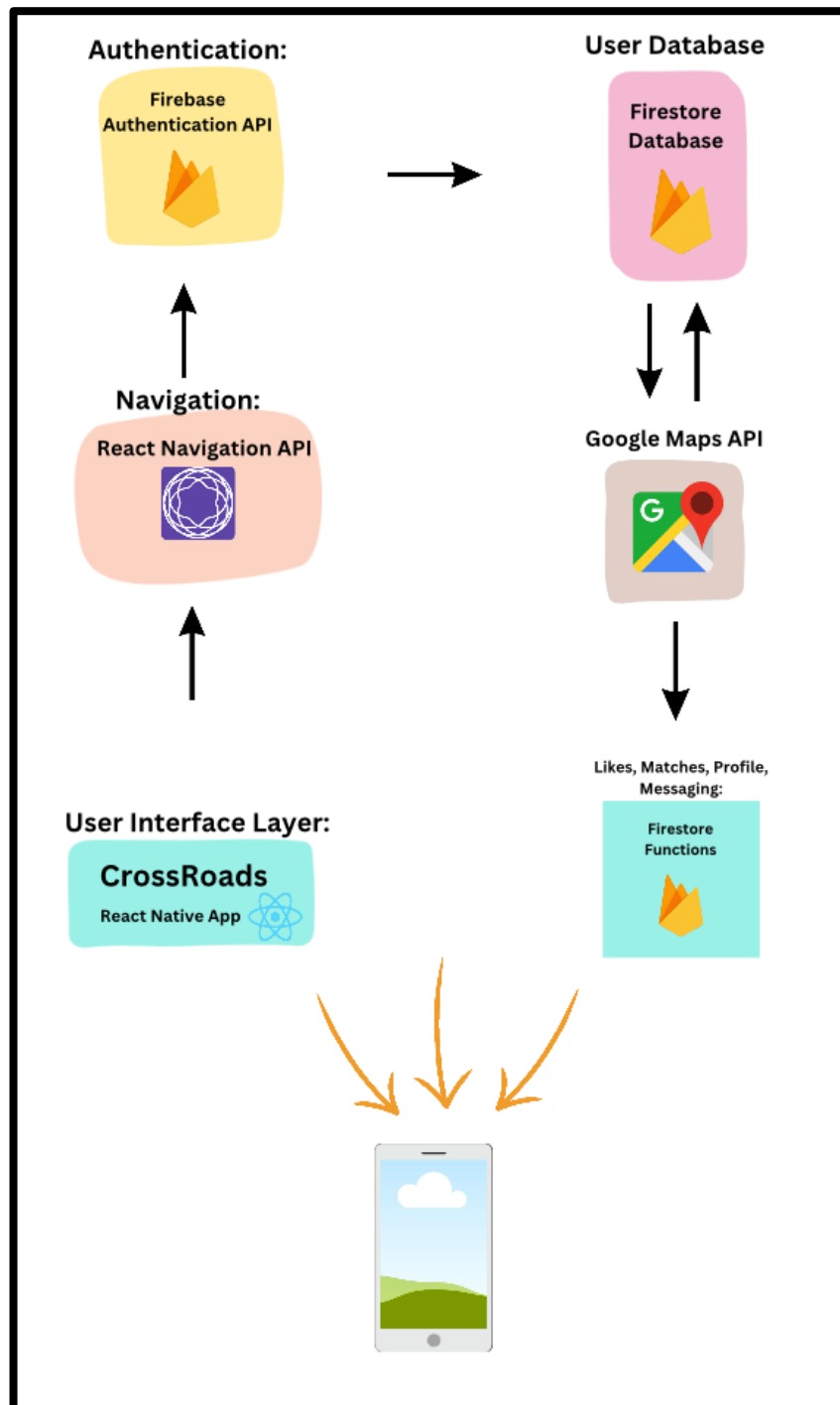
If an error occurs in the snapshot listener, it logs the error and shows an alert.

7. Social Media Content Upload and Browsing (Core) [Completed]

Priding ourselves on being a dating app with social media-like functionalities, CrossRoads also allows users to post media content about their daily activities, and interactions with people or simply just browse media content curated on the platform. This encourages more organic connections where users are able to let their matches know more about their daily lives in ways apart from texting.



Software Architecture



Tech Stack

1. Expo Go - Frontend framework
2. ReactNative (JS) - Frontend framework
3. Firebase & Firestore (Backend framework - Account authentication, messaging, storage)
4. Figma - UI/UX
5. Git & Github - Version Control

Software Engineering Practices

1. SOLID principles

Use of modular components and functions to make it easier to extend functionality (e.g. adding new components, new screens)

Each component demonstrates single responsibility, although they can be further improved if they were broken down into different functionalities.

2. Version Control

Pull Requests

3. Two-week Sprints

Our work schedule is organized into two-week sprints, a common practice in the industry. Every two weeks, we hold a meeting to review the progress made, assess adherence to the timeline, and discuss our work plan; this approach helps us meet deadlines while allowing us the flexibility to adjust our development plans in response to any unexpected issues that may arise.

4. Security Measures

Concealing Google Access Token

To enhance security for our Firebase Authentication and Firestore database, private keys and tokens are stored in a .env file. This file is listed in .gitignore, ensuring it is not uploaded to the remote master branch during commits.

Consequently, this prevents unauthorized modifications and access to the database through our GitHub Repository.

Challenges Faced

1. Inefficient testing; having to create multiple accounts to log in and out
2. Firebase and Expo Go setup was quite confusing and time-consuming
3. Firestore Usage - Limited queries
4. Deployment for milestone 2 was not as straightforward as milestone 1 due to the use of Firestore (handling Google Services JSON file, etc)
5. Messy information storage; Retrieving inaccurate information from wrong collections
6. Testing of Geolocation feature
7. Experimenting with backend rules and indexes

Areas for improvement

1. Security
2. Interface Design & User Friendliness
3. More centralised profile information handling

Proof-of-Concept

Video Demonstration Link:

[6525 Demo.mp4](#)

Test Accounts:

1. zhang@gmail.com
Password
2. calvin@calamaribasil.net
Password

User Testing

[CrossRoads - Tests Conducted](#)

Project Log Link

[Project Log.xlsx](#)